Build Your Own AIOps Agent Using MCP

Technical Insights into MCP, AIOps, and Local AI Agents

About Me

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Agenda

1. Introduction and Overview
2. Understanding AIOps
3. MCP (Model Context Protocol)
4. Architecture
5. Building a Real-World Applications
6. DEMO
7. Takeaways

Introduction and Overview

Build AIOps Agent Using MCP

Limitations of Manual Monitoring

Prone to human error and slow response times

Fragmented Tools and Scattered Logs

Multiple systems complicate analysis and troubleshooting

AI and Automation for AIOps

Enables proactive, unified, and intelligent infrastructure management

Understanding MCP

Understanding Model Context Protocol

MCP Protocol Overview

Defines standard for LLM-to-tool interactions using **JSON-RPC** over stdio. Act as Universal standard (like **USB-C** type)

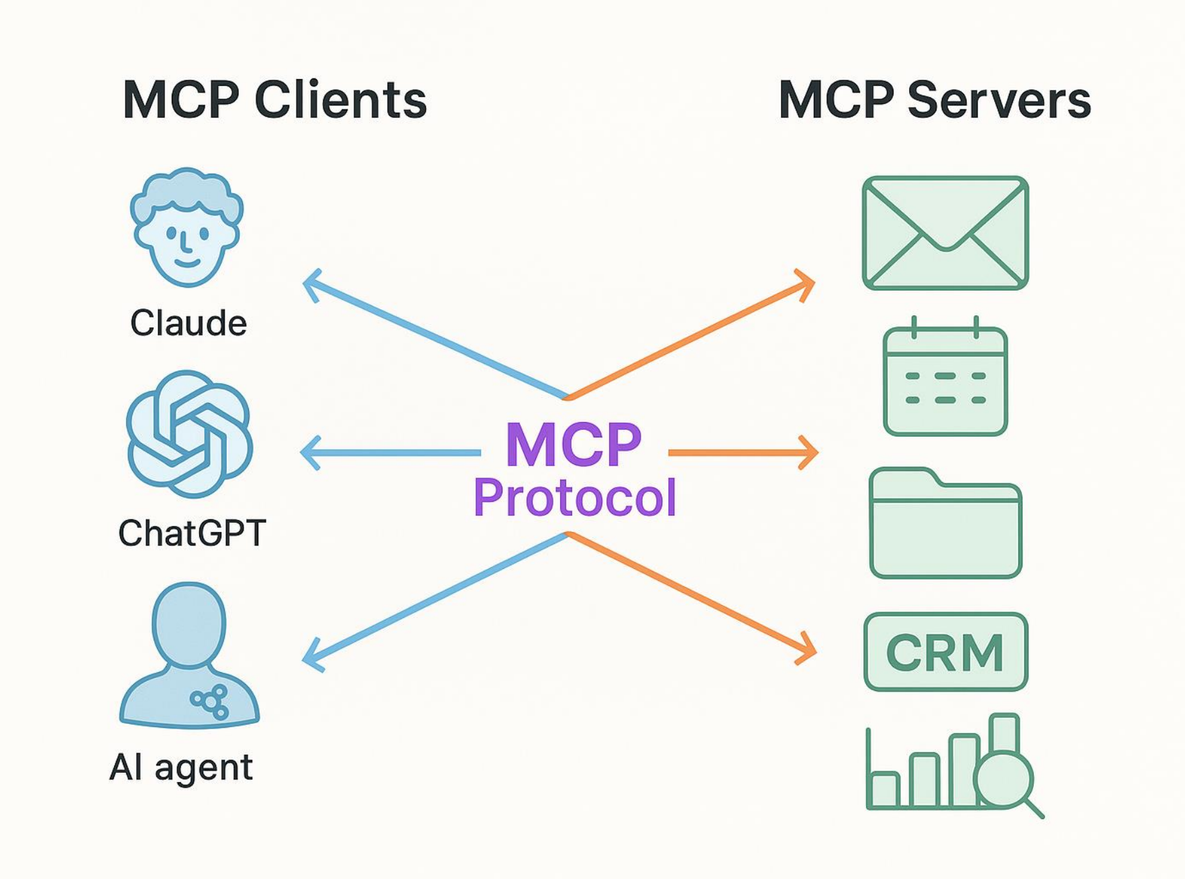
Benefits for MCP

Turns "**Passive Chatbots**" into "**Active Agents**".

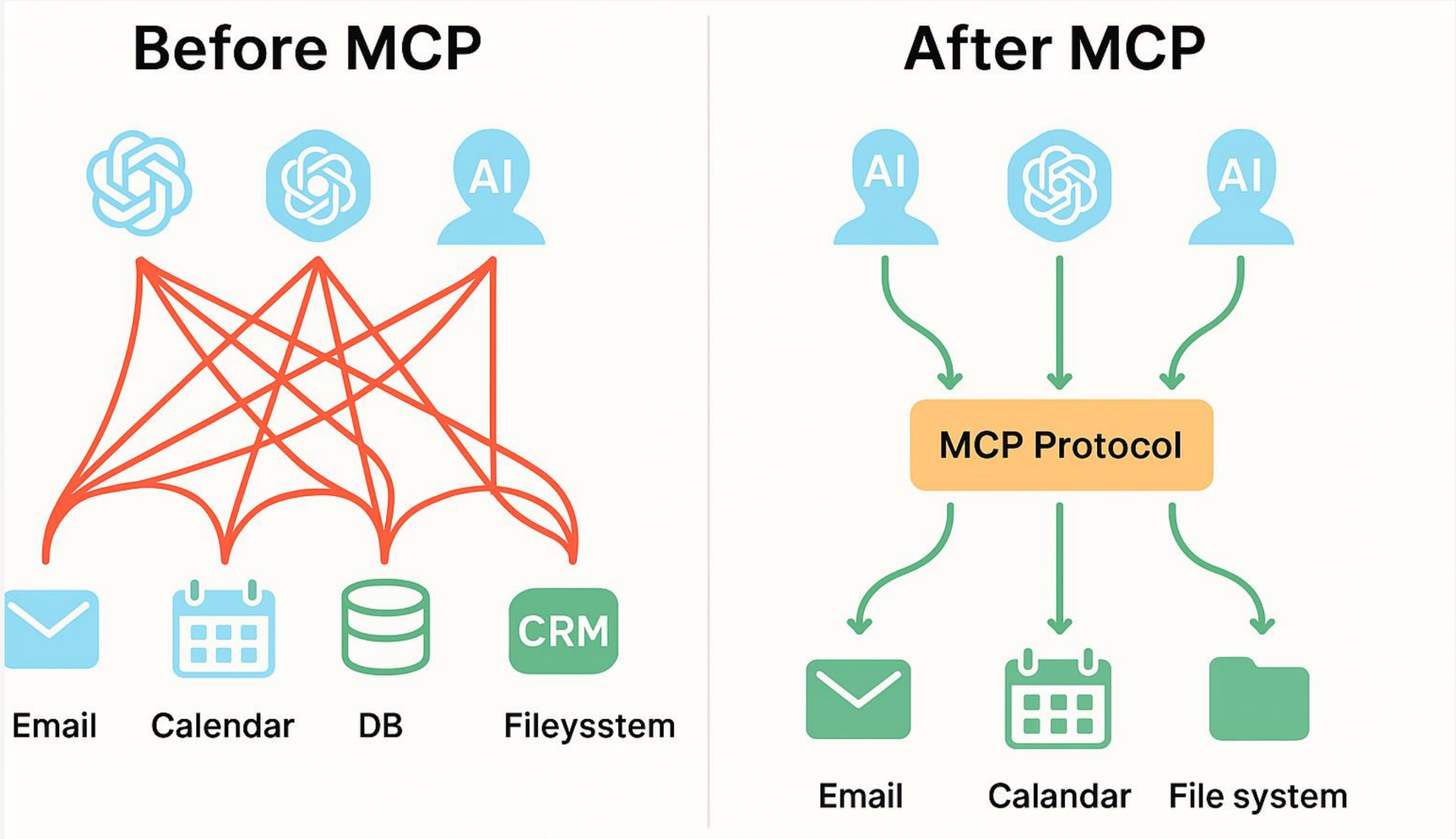
Reduces Hallucinations, Security & Governance.

Structured and Safe Communication

Ensures deterministic, auditable, and secure tool usage by language models.



Understanding Model Context Protocol



Building and Extending MCP Servers

python Copy

```python
# server.py (very small MCP-style JSON-RPC handler)
import sys, json

def handle_request(req):
    method = req.get("method")
    if method == "get_system_info":
        return {"cpu": 12.5, "mem": 45.1}
    return {"error": "unknown method"}

# read JSON-RPC from stdin (MCP uses stdio)
for line in sys.stdin:
    req = json.loads(line)
    res = {"id": req.get("id"), "result": handle_request(req.get("params", {}))}
    print(json.dumps(res), flush=True)
```

Example JSON-RPC request & response

Request

json Copy

```
{"id":1,"method":"get_system_info","params":{}}
```

Response

json Copy

```
{"id":1,"result":{"cpu":12.5,"mem":45.1}}
```

Understanding AIOps

AIOps: Automating Operations with AI

AI Integration in IT Operations

MCP enables structured, safe tool calls for LLMs

Automated Monitoring and Analysis

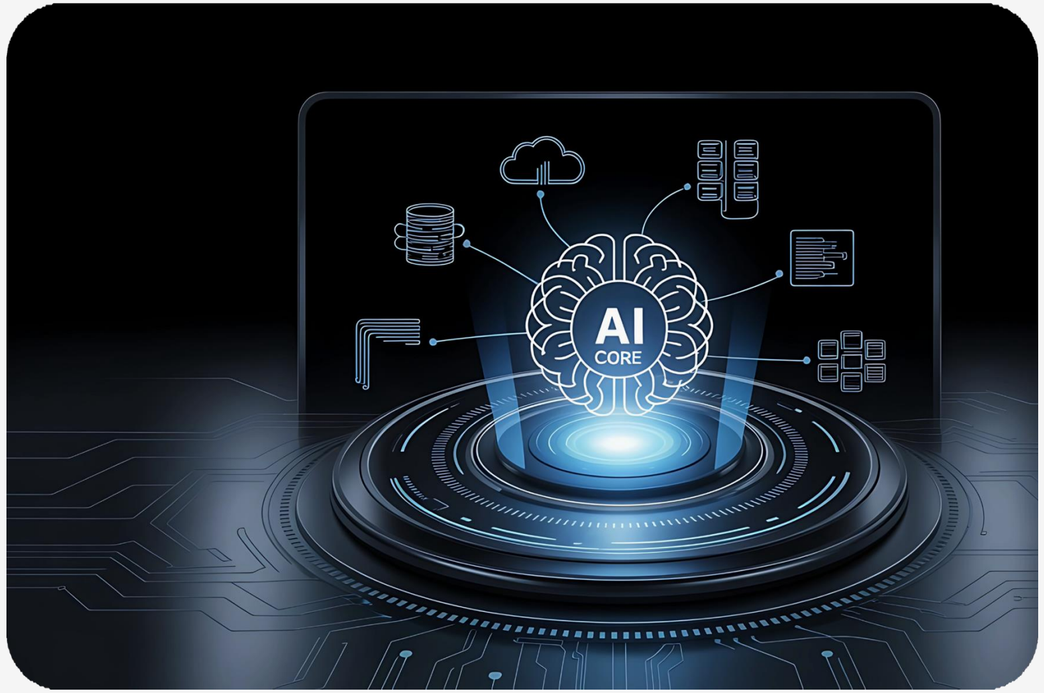
InfraGPT-MCP automates log analysis, health checks, and threat detection

On-Premises AIOps Agent

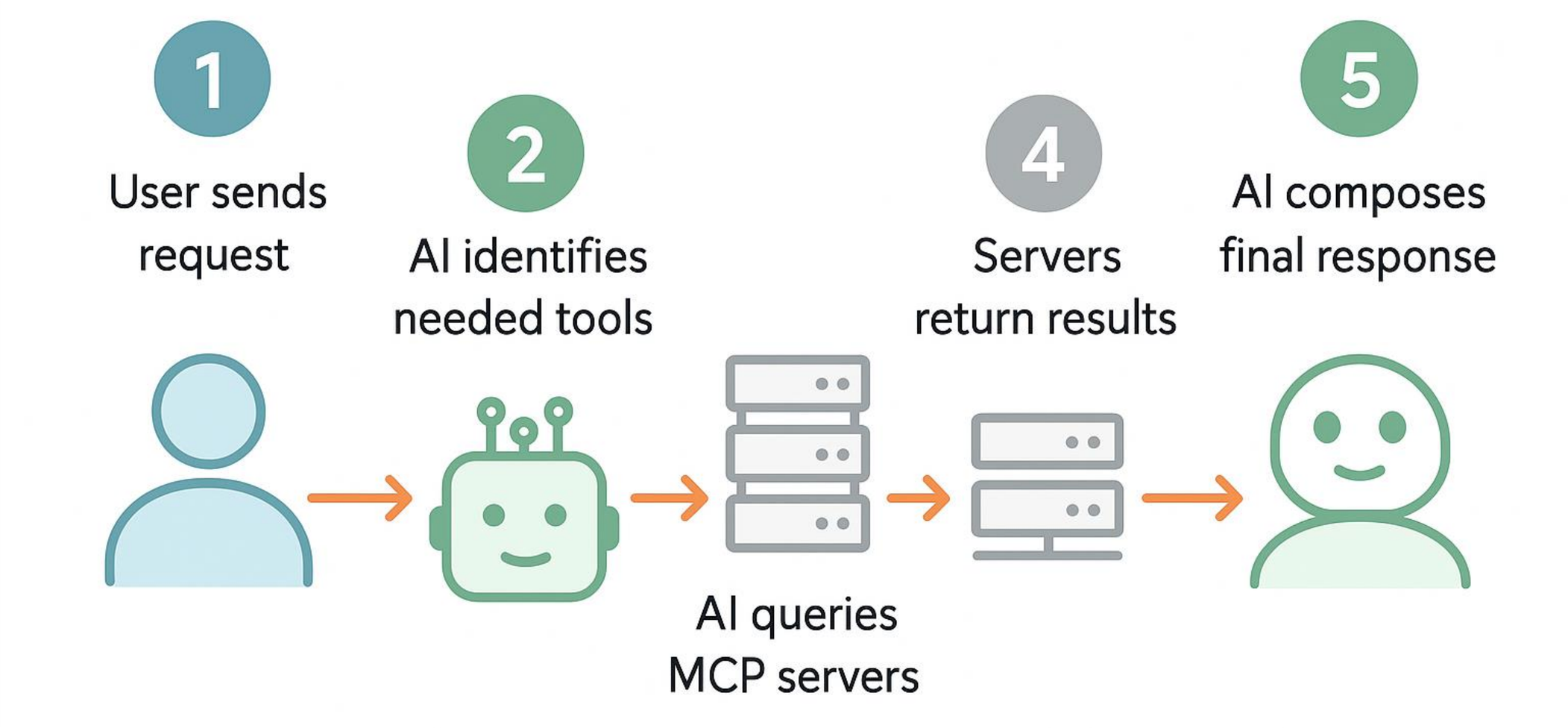
Runs locally with Ollama, ensuring privacy and zero cloud dependency

Practical Implementation

InfraGPT-MCP delivers real-time insights and recommendations on laptops



AIOps: Automating Operations with AI



What Is InfraGPT-MCP?

AI-Powered Infrastructure Monitoring

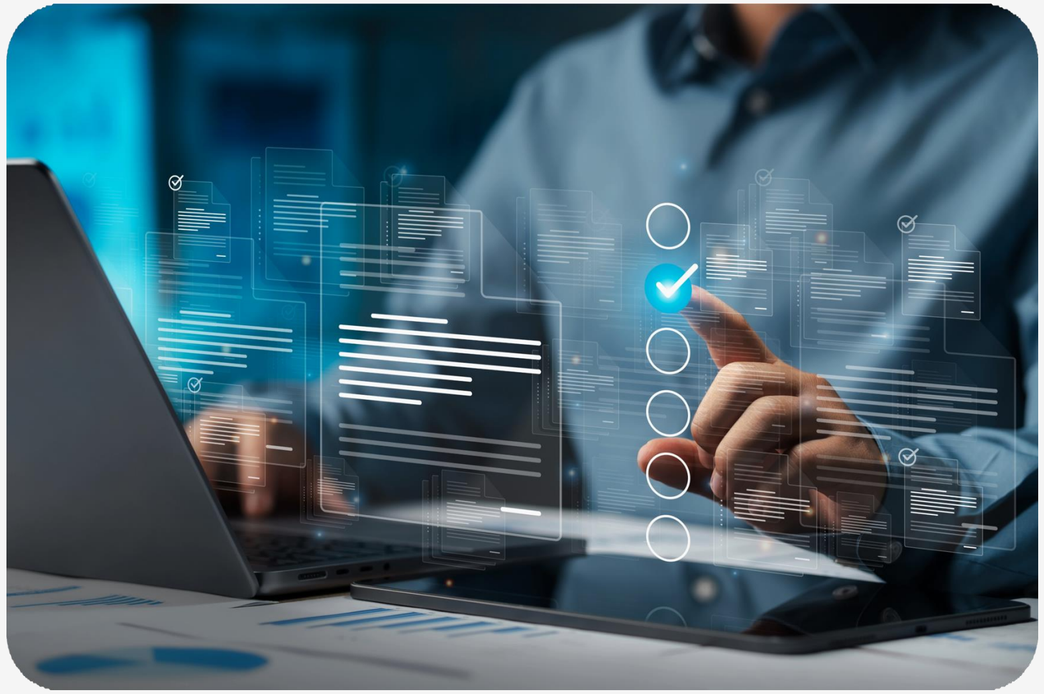
Automates system checks and log analysis using AI.

Model Context Protocol (MCP) Foundation

Enables structured, safe tool calls via JSON-RPC.

Privacy-First, Local Deployment

Runs on-premise with no cloud dependency using local LLMs.



InfraGPT-MCP Architecture Deep Dive

LLM Client Integration

Claude Desktop or other LLM clients connect to the MCP server for requests.

MCP Server as Middleware

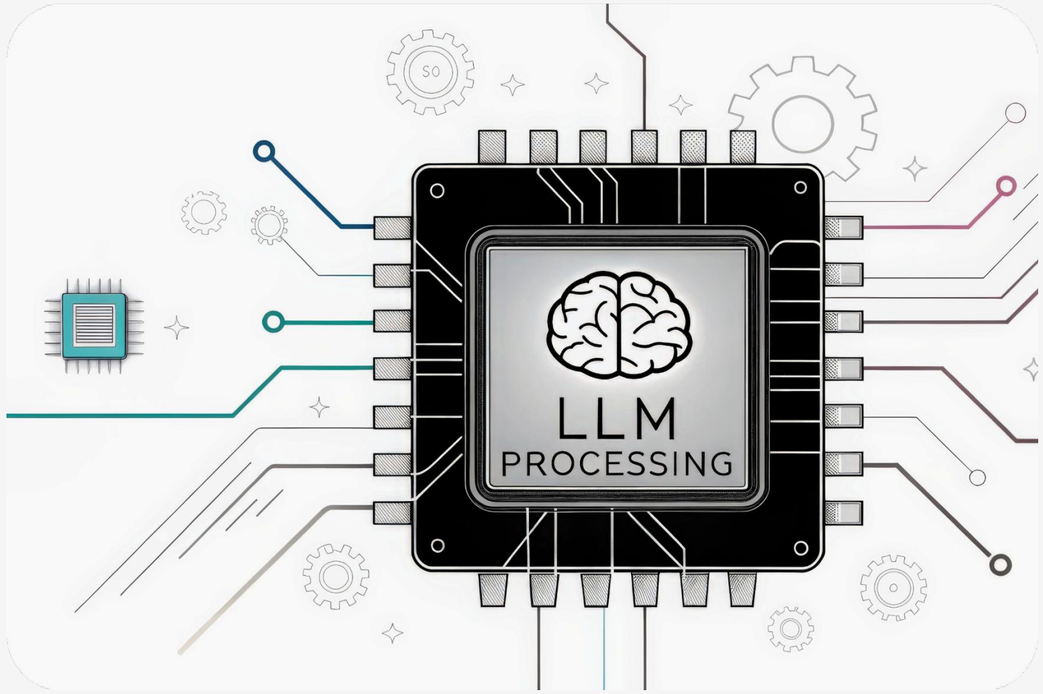
MCP server receives requests and interfaces with system tools using JSON-RPC.

System Tools Connectivity

Utilizes psutil for system metrics and log readers for log analysis.

Optional Local AI Analysis

Gemini or Ollama can be integrated for on-prem AI-powered analysis of system data.



Key Features and Capabilities

Comprehensive Toolset

Own integrated tools:

system info, service status, logs, network, user info, health check.

Conversational Monitoring

Enables real-time, interactive infrastructure insights via natural language queries.

Privacy and Local Processing

Mock mode and 100% local data handling ensure privacy and offline capability.

Real-World Applications and Takeaways

Real-World Use Cases and Demos

System Health Scoring

- Generates health scores and metrics for rapid infrastructure assessment.
- Provides actionable recommendations for system optimization.

Security Log Analysis

- Analyzes login events and system logs to detect potential threats.
- Summarizes security issues for proactive incident response.

Performance Monitoring

- Identifies top CPU-consuming processes for performance tuning.
- Delivers real-time insights to support resource management.

Key Takeaways for Practitioners

Safe Integration of LLMs and System Tools

MCP enables structured, secure tool calls from LLMs using JSON-RPC.

Practical AIOps on Local Machines

InfraGPT-MCP delivers real-time monitoring and analysis directly on your laptop.

Open, Extensible, and Privacy-First Design

Supports on-prem LLMs, mock mode, and easy tool extension for technical teams.

Questions ?

Thank You !

LinkedIn

https://www.linkedin.com/in/shivkumarsah

Github Code & Slides

https://github.com/shivkumarsah/InfraGPT-MCP